



Paper ID : 250582

Printed Page: 1 of 2
Subject Code: RCAI802

Roll No:

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MCA-INT
(SEM VIII) THEORY EXAMINATION 2024-25
ADVANCED JAVA PROGRAMMING

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

2 x 7 = 14

1. Attempt *all* questions in brief.

a.	What do you mean by Java Beans?
b.	Discuss Java message service and API?
c.	What does it mean by JNDI?
d.	What are the major tiers in the J2EE architecture?
e.	What does the parseInt() function do?
f.	Why is JSP preferred for dynamic web page development?
g.	How do servlets differ from JSP?

SECTION B

7 x 3 = 21

2. Attempt any *three* of the following:

a.	How does JMS Architecture work? Explain its functionality with appropriate example?
b.	Explain the MVC(Model-View-Controller) architecture and also explain the directory structure of a web application that follows this architecture?
c.	Discuss the characteristics of JavaScript? Explain the use of JavaScript dialog boxes and their role in user interaction?
d.	What is Tomcat server? What are the stages of the JSP life cycle? Explain in detail?
e.	What are the challenges of web application? Explain how Java Servlets handle HTTP requests and responses in a Java EE environment?

SECTION C

7 x 1 = 7

3. Attempt any *one* part of the following:

a.	What are the Advantages of Enterprises Java Beans? Differentiate Stateless session bean and Stateful session bean with proper example?
b.	Explain the steps of writing JMS clients for sending and receiving messages in details?

4. Attempt any *one* part of the following:

7 x 1 = 7

a.	What is J2EE container? Discuss the various types of it? Also discuss J2EE Architecture with suitable diagram?
b.	Why is java mail API used in Java enterprises? Describe the steps to send an email using java mail API?



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5. Attempt any one part of the following:

7 x 1 = 7

a.	Describe how variables are declared in JavaScript? How do event handlers work in JavaScript?
b.	Explain the functions of JavaScript built-in objects such as mathematical computations, date-time manipulation with appropriate examples?

6. Attempt any one part of the following:

7 x 1 = 7

a.	Discuss implicit and explicit objects in jsp? How do JSPs differ from servlets in terms of design and usage?
b.	How can variables and methods be declared and used within JSP pages? Explain with examples?

7. Attempt any one part of the following:

7 x 1 = 7

a.	What is JDBC? How to access database using JDBC? Explain the steps with appropriate example?
b.	What do you mean by session tracking? Explain various mechanisms used for session tracking in servlets?